

MASS-ENG-006

Workflow Engine Specification

Page 1 – Purpose & Scope

Purpose

Define the orchestration engine responsible for coordinating business processes across MASS HQ.

Objectives

- Coordinate long-running processes
- Execute state-based workflows
- Consume and publish enterprise events
- Support human and automated tasks

Out of Scope

Business rules unique to individual modules and user interface logic.

Page 2 – Architecture

Core Components

Workflow Definition Registry, Execution Engine, State Manager, Task Queue, Rules Evaluator Interface, Event Integration Layer.

Relationships

Consumes the Event Bus and Enterprise Core services while orchestrating Identity, Documents, Knowledge, AI, and Notifications.

Page 3 – Functional Specification

Requirements

1. Register workflow definitions.
2. Execute workflow instances.
3. Maintain workflow state.
4. Support conditional branching and approvals.
5. Pause, resume, retry, and terminate workflows.
6. Emit workflow lifecycle events.

Non-Functional

Durable execution, idempotent processing, scalability, auditability, recoverability.

Page 4 – Interfaces & Examples

Interfaces

StartWorkflow(), CompleteTask(), AdvanceState(), CancelWorkflow(), GetStatus().

Example Flow

New Case Created → Event Bus → Workflow Engine starts onboarding workflow → Tasks assigned → Approvals completed → Notification sent → Workflow completed.

Page 5 – Construction Package

Claude Checklist

Implement workflow model; state machine; execution engine; task queue; persistence; event integration; workflow APIs; monitoring; unit and integration tests.

Definition of Done

Workflow instances execute deterministically, survive restarts, integrate through the Event Bus, and expose auditable execution history.

Constitution References

Enterprise Governance, Platform Principles, Knowledge, Security, Enterprise Structure.